

CH-8

HOW DO ORGANISMS REPRODUCE

CLASS 10

BIOLOGY

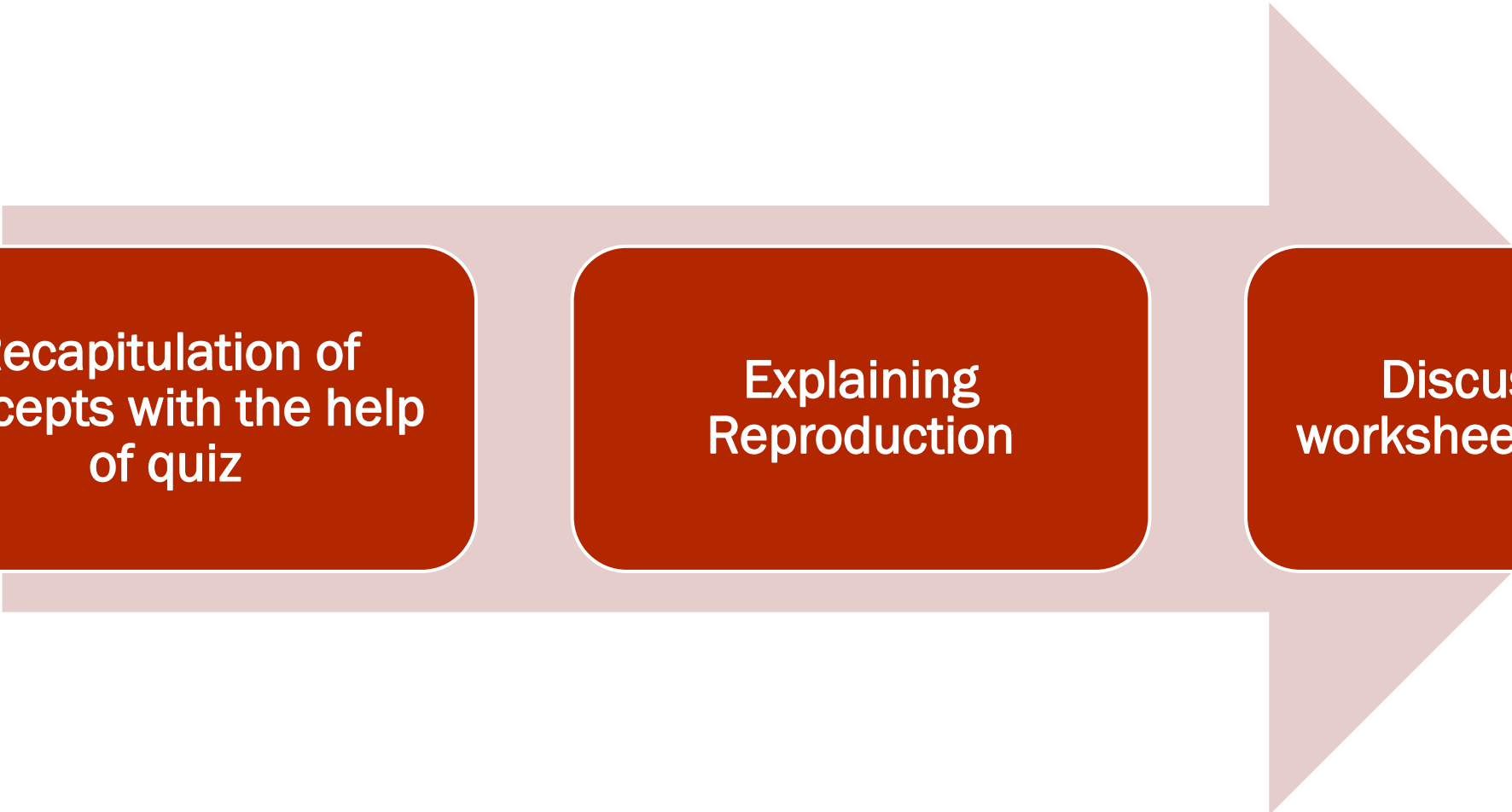
SESSION- 1

MS. PARIDHI AGARWAL

Live Class Etiquettes

- Keep your camera and mic on mute mode.
- Teacher will give you time to put up your queries/questions.
- Use the chat window to put up queries/questions ONLY when teacher asks you to do so. Do not use it for greetings or making other irrelevant comments.
- You can also use the option of raise your hand to put up a question, when teacher asks you to do so.
- When teacher asks you a question, switch on your video and audio to answer.
- All resources used during the session will be posted under files tab in your team.

PLAN OF THE DAY



Recapitulation of
concepts with the help
of quiz

Explaining
Reproduction

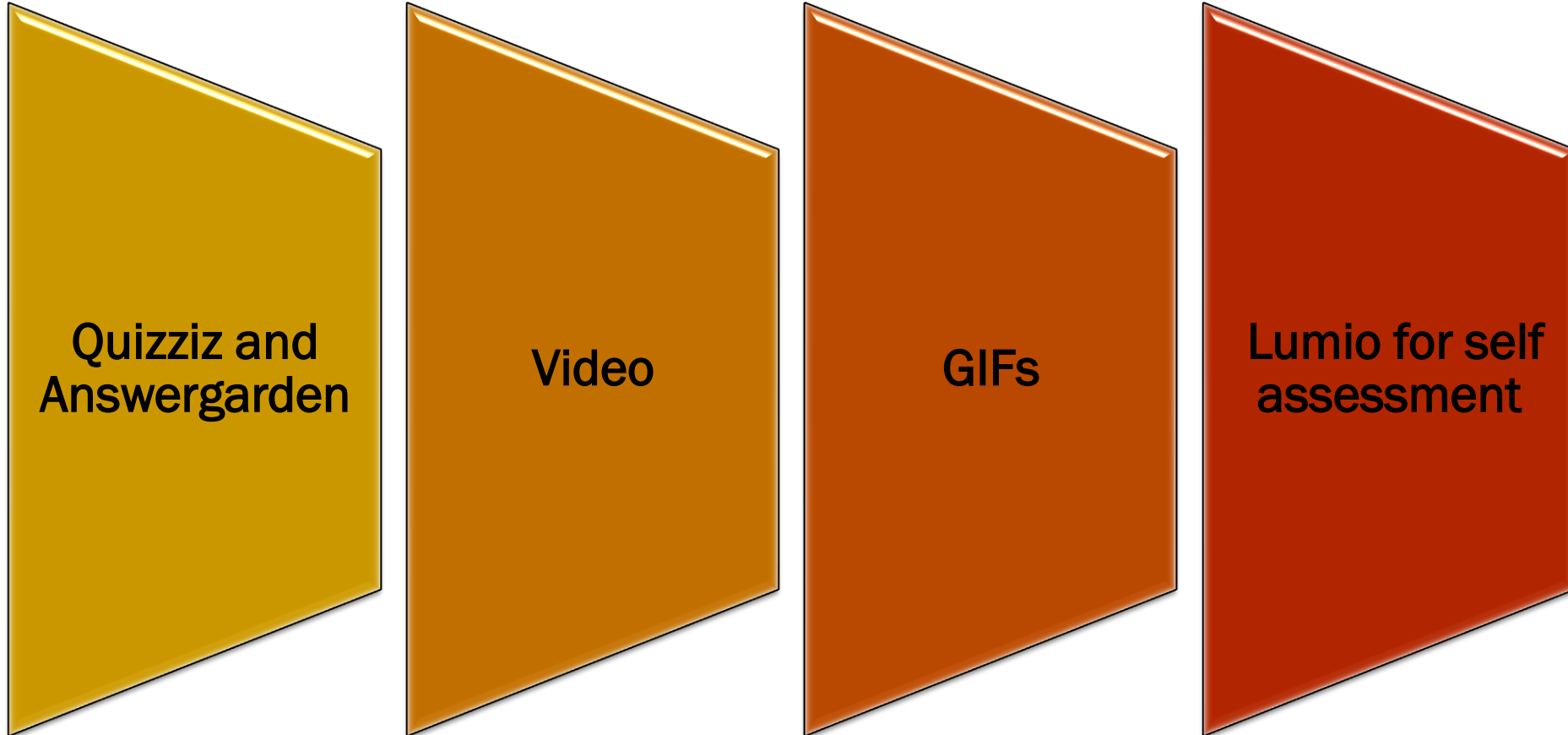
Discussion of
worksheet questions

LEARNING OBJECTIVES



Define Reproduction
Classify Reproduction into various types
Analyze the process of Reproduction
Answer the questions based on topics taught

TOOLS AND TECHNIQUES USED



PREVIOUS KNOWLEDGE

Meaning of Reproduction

Types of Reproduction



<https://quizizz.com/admin/quiz/613ff38747c0ea001d6f4ec1/reproduction>

LIFE PROCESSES

NUTRITION

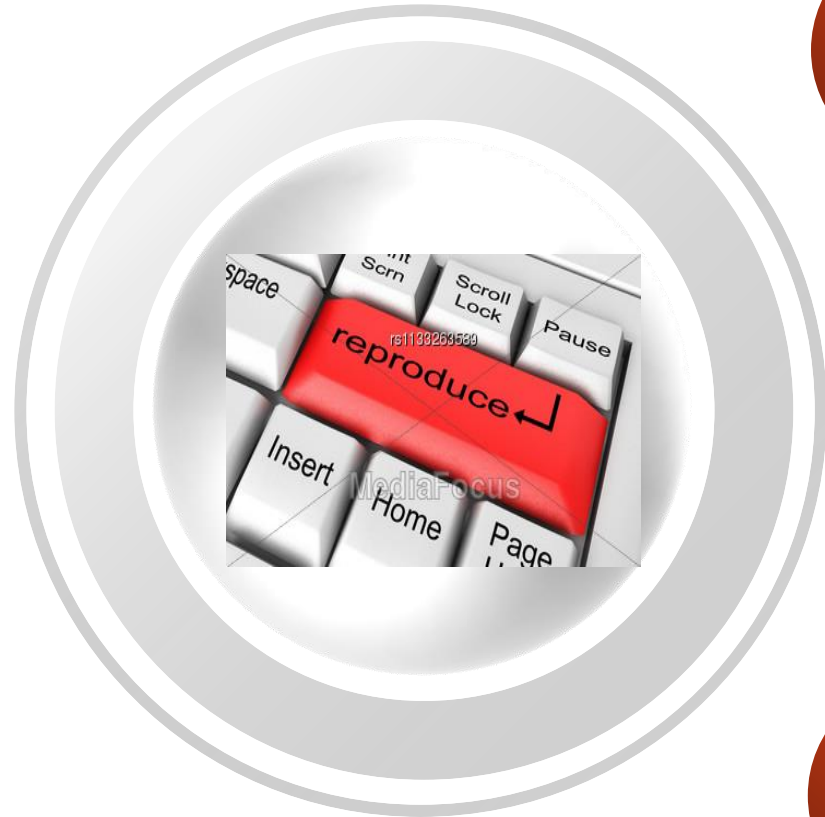
RESPIRATION

TRANSPORTATION

EXCRETION

REPRODUCTION

Why do Organisms Reproduce ?



1

It is not essential for survival of organisms

2

It utilizes energy

3

Time taking, long and complicated process...

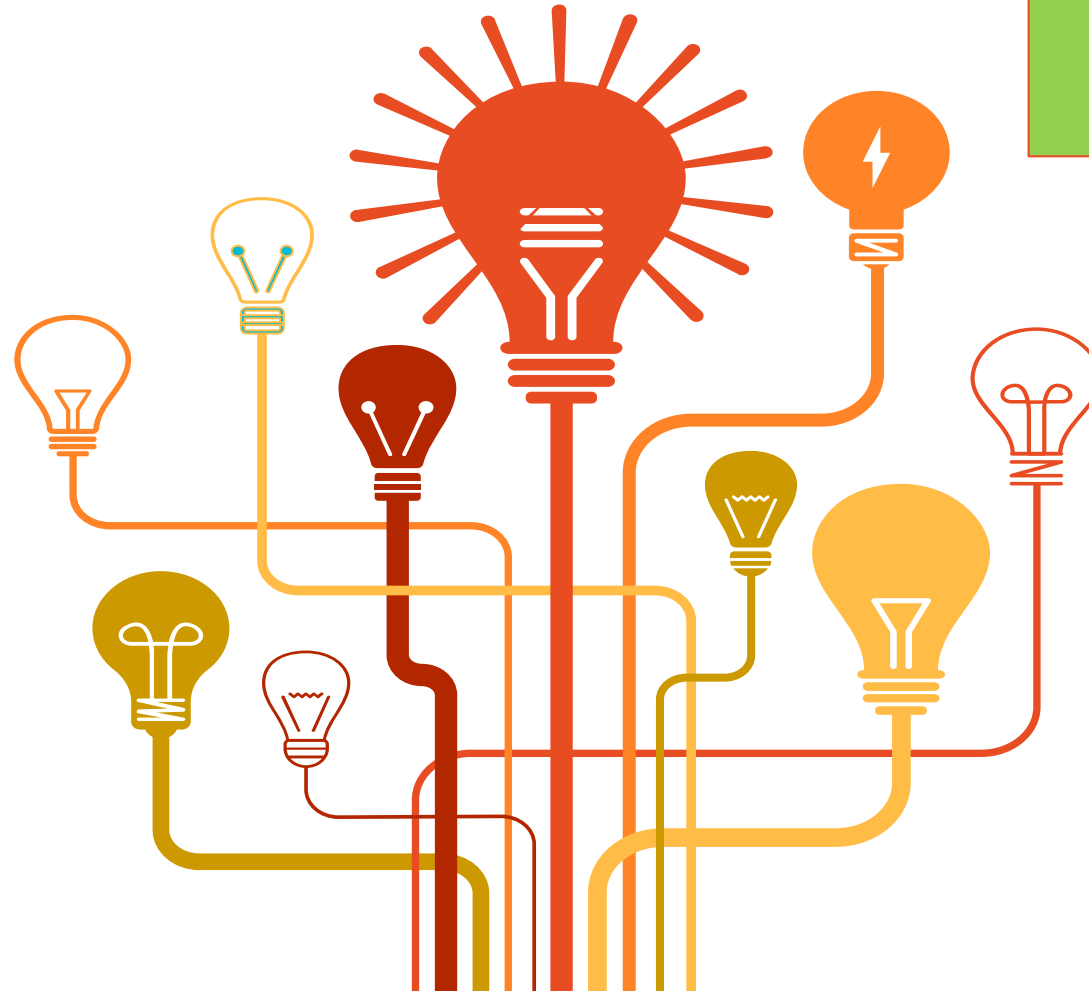
Why do Organisms Reproduce ?

To continue the
chain of life

For survival of
the species

Their existence
can be felt

To produce
organisms of
their own kind



SAME V/S DIFFERENT



<https://answergarden.ch/2108121>

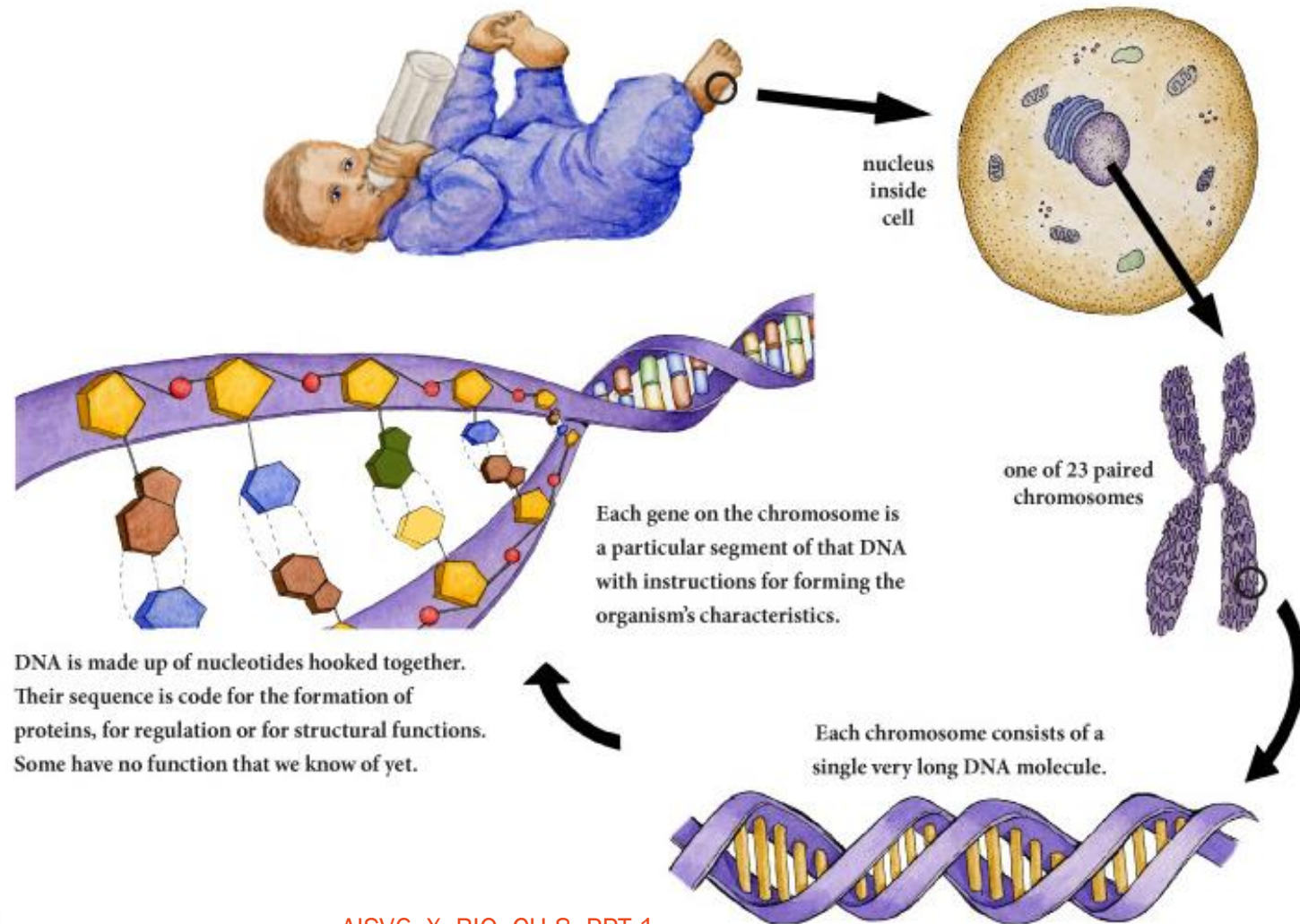
<https://answergarden.ch/2108122>

DO ORGANISMS CREATE EXACT COPIES OF THEMSELVES?

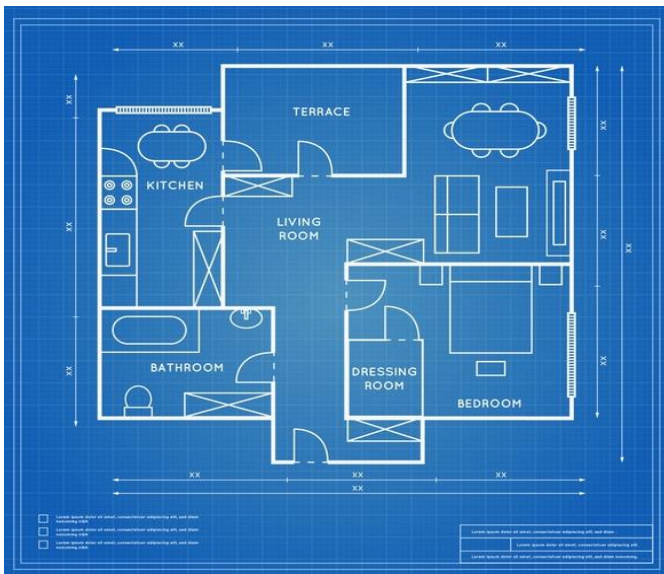


Organisms look similar because their body designs are similar

Where is our Genetic Information?



Blueprint



Blue Print

S. No.	Form of Questions	VSA	SA-I	SA-II	LA	MCQ	Total
UNITS							
1.	Chemical Substances- Nature and behaviour						
	a) Carbon Compounds	1(1)	—	3(1)	5(1)	5(5)	14 } 21(11)
	b) Periodic classification of elements	—	4(2)	3(1)	—	—	
2.	World of Living						
	a) Reproduction	—	4(2)	3(1)	5(1)	6(6)	18 } 27(13)
	b) Heridity and evolution	—	—	9(3)	—	—	
3.	Natural Phenomenon						
	a) Reflection and refraction of light	—	2(1)	6(2)	5(1)	5(5)	18 } 26(13)
	b) Human eye and the colourful world	1(1)	4(2)	3(1)	—	—	
4.	Natural Resources						
	a) Fossil Fuels	—	2(1)	—	—	—	2 } 6(4)
	b) Management of natural resources, The regional environment	—	2(1)	—	—	—	
	c) Our environment	2(2)	—	—	—	—	2
		4(4)	18(9)	27(9)	15(3)	16(16)	80 (41)

01

Organisms look similar because their body designs are similar

02

Thus blueprints for these designs should be similar

03

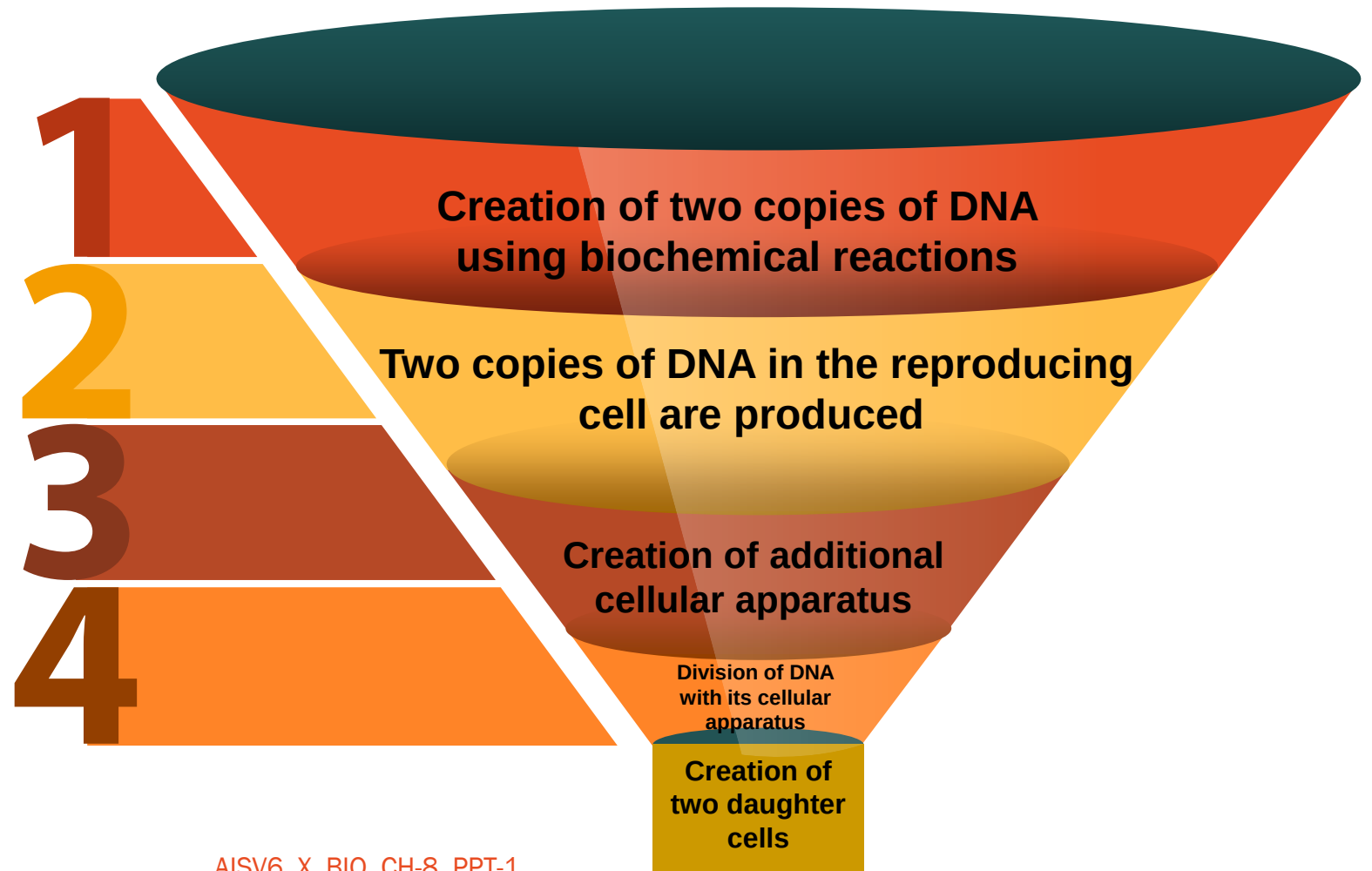
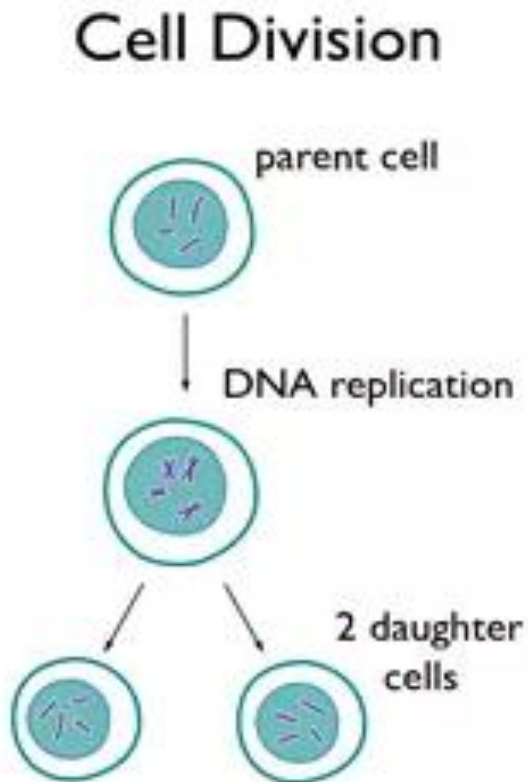
Making copies of the blueprints of body design and transferring to the next generation

04

The blueprint of body design is stored in DNA

- ✓ Chromosomes in the nucleus of a cell contain information for inheritance of features from parents to next generation in the form of DNA (Deoxyribo Nucleic Acid) molecules.
- ✓ The DNA in the cell nucleus is the information source for making proteins.
- ✓ If the information is changed, different proteins will be made.
- ✓ Different proteins will eventually lead to altered body designs.

DNA Copying/ DNA Replication



DNA Copying/ DNA Replication

1
Step

Unwinding of Double stranded DNA Helix



2
Step

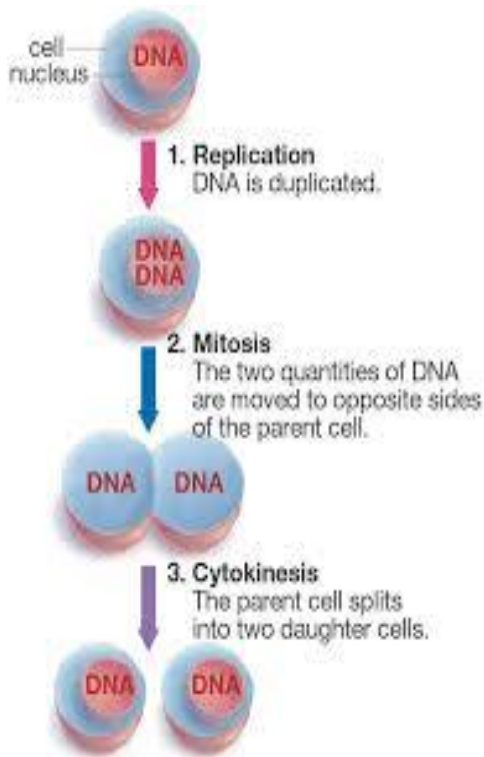
Formation of new strand on parental strand

3
Step

Utilisation of Chemicals found in the nucleus

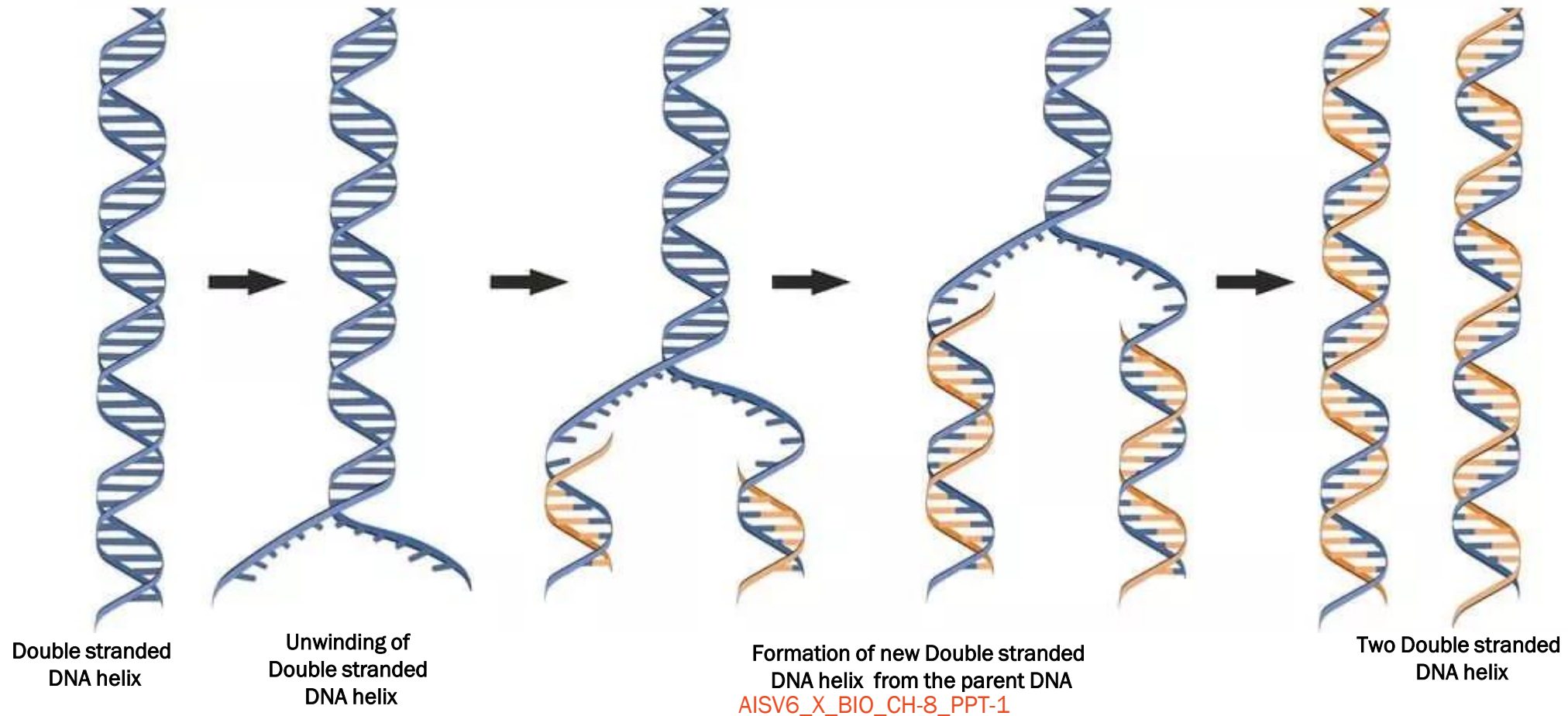
4
Step

Formation of 2 Double Stranded DNA Helix



DNA Copying/ DNA Replication

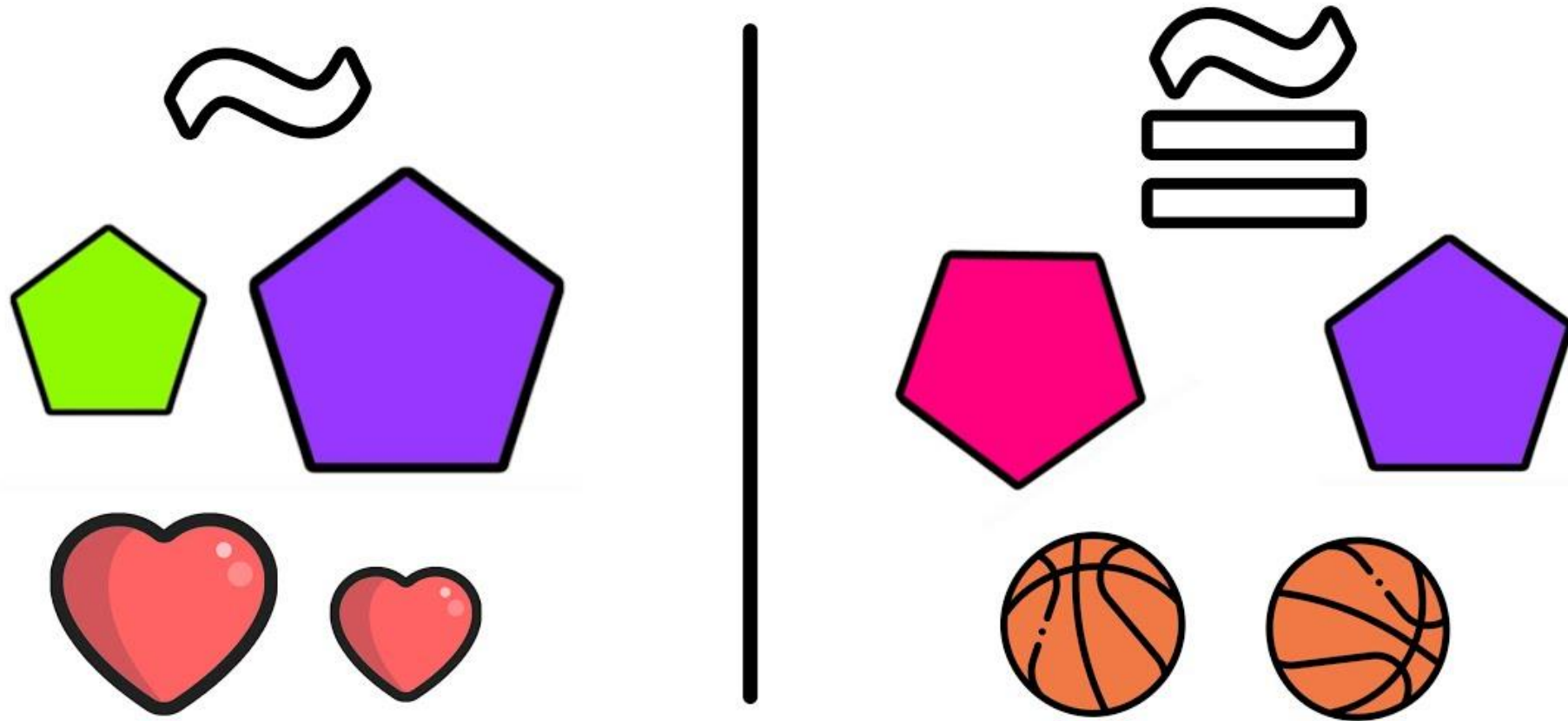
DRAW IN
NOTEBOOK



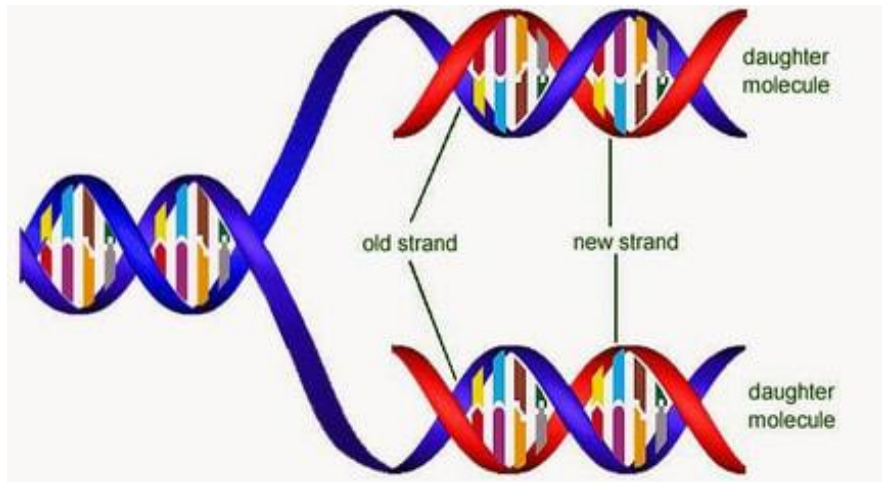
DNA is the information source for making proteins

Different proteins leads to different body designs

SIMILAR VS CONGRUENT



These two cells are of course similar, but are they likely to be absolutely identical?



❑ It will depend on how accurately the copying reactions involved occur.

❑ No bio-chemical reaction is absolutely reliable. Therefore, it is only to be expected that the process of copying the DNA will have some variations each time.

❑ As a result, the DNA copies generated will be similar, but may not be identical to the original.

❑ These are called as Variation.

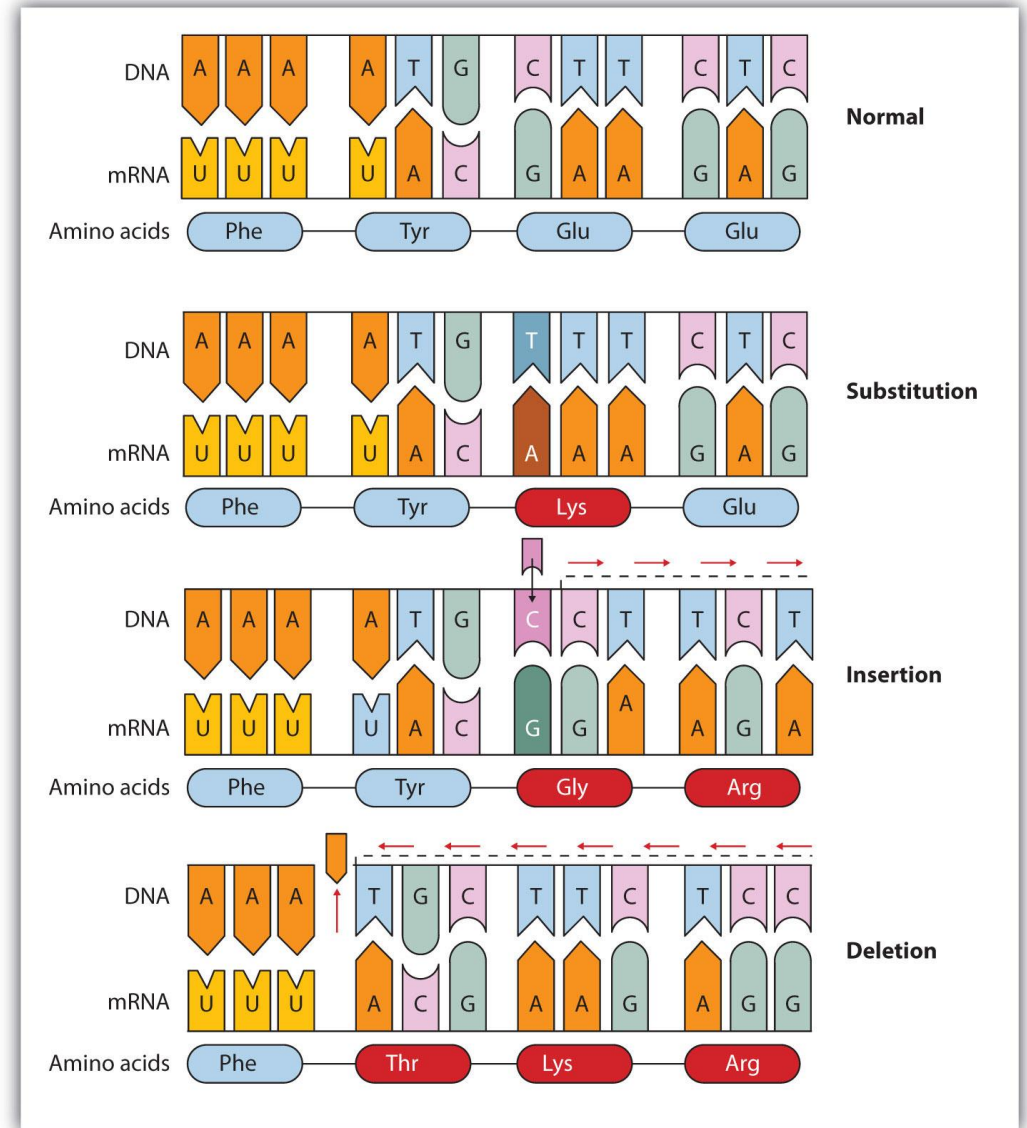
Variations

❑ Some of these variations might be so drastic that the new DNA copy cannot work with the cellular apparatus it inherits. Such a newborn cell will simply die.

❑ On the other hand, there could still be many other variations in the DNA copies that would not lead to such a drastic outcome.

❑ Thus, the surviving cells are similar to, but subtly different from each other.

This inbuilt tendency for variation during reproduction is the basis for evolution

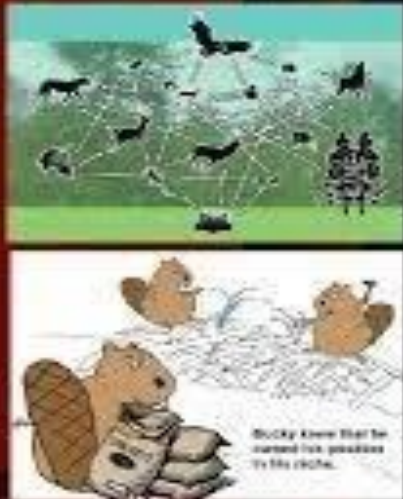


What are NICHES?

What are Niches?

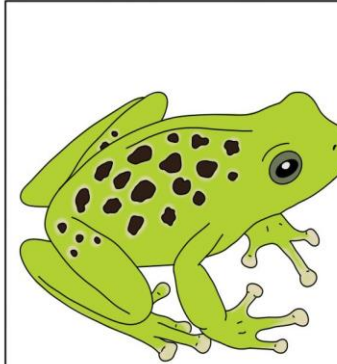
- **Niche** is the **role of an organism** within a particular ecosystem. **Includes:**

1. What it eats.
2. What eats it.
3. Its habitat.
4. Nesting site, range and habits.
5. What effect it has on the other populations.
6. What effect it has on the environment.



NICHE EXAMPLE:

BARKING TREE FROG



NICHE

Habitat: wooded regions, wetlands

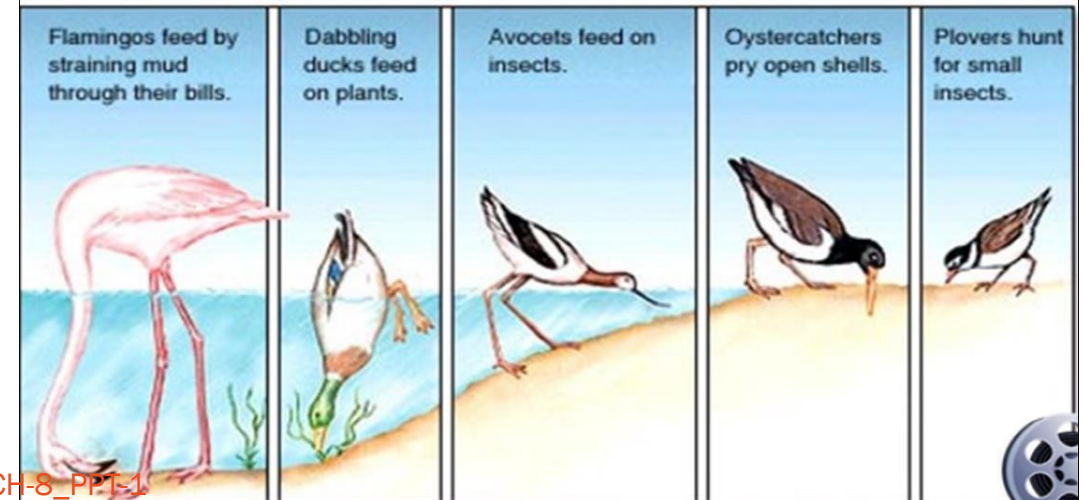
Behaviors: often found in trees, can burrow, barking call attracts mates

Feed on: small insects

Eaten by: birds, snakes, raccoons

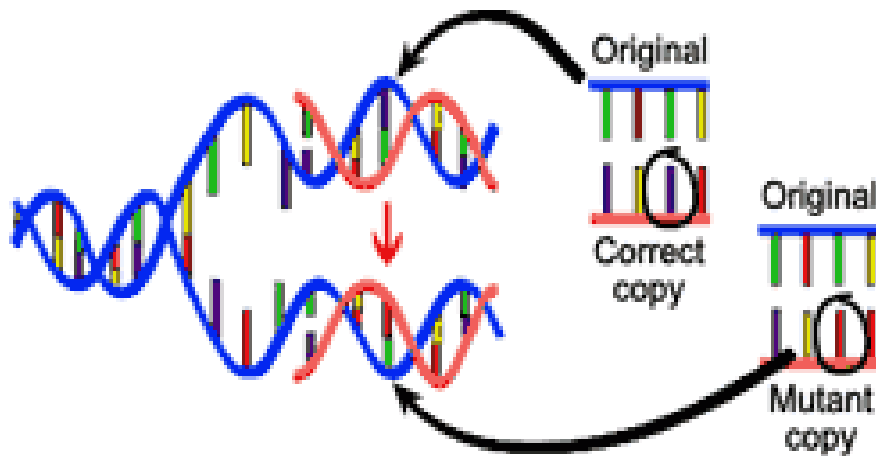
Impact on ecosystem: limit insects, aerate soil, serve as food source

An organism's **habitat** is its "address" while its **niche** is its "occupation"



Importance of Variations

- Populations of organisms fill well-defined places, or niches, in the ecosystem, using their ability to reproduce.
- The consistency of DNA copying during reproduction is important for the maintenance of body design features that allow the organism to use that particular niche.
- *Reproduction is therefore linked to the stability of populations of species.*



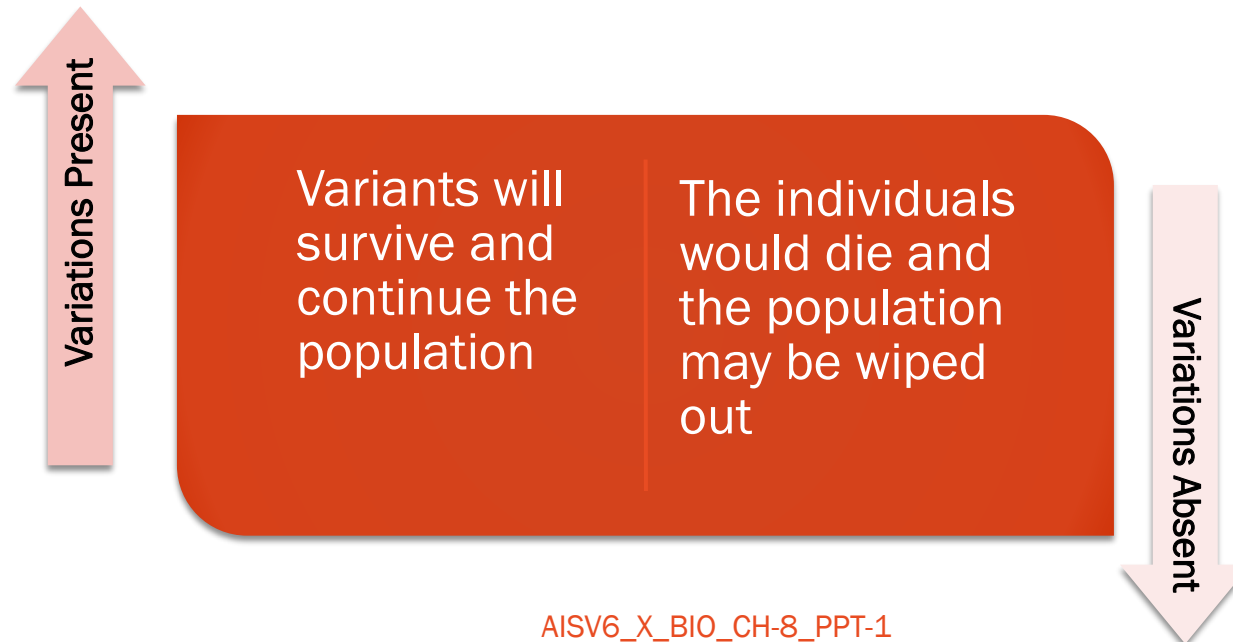
More Variations

- Less consistency
- Less stability of the population

Less Variations

More consistency
More stability of the population

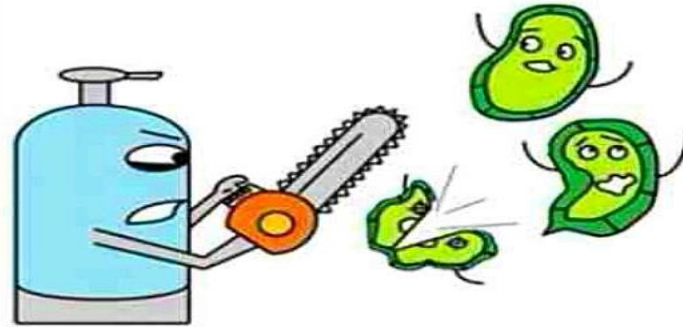
- ❑ However, niches can change because of reasons beyond the control of the organisms. Temperatures on earth can go up or down, water levels can vary, or there could be meteorite hits, etc.
- ❑ If a population of reproducing organisms were suited to a particular niche and if the niche were drastically altered, the population could be wiped out.
- ❑ However, if some variations were to be present in a few individuals in these populations, there would be some chance for them to survive.



I AM
HAND SANITIZER!!!



I KILL 99.99% OF GERMS



I SPARE THAT .01%



SO THAT IT CAN
DELIVER A MESSAGE...



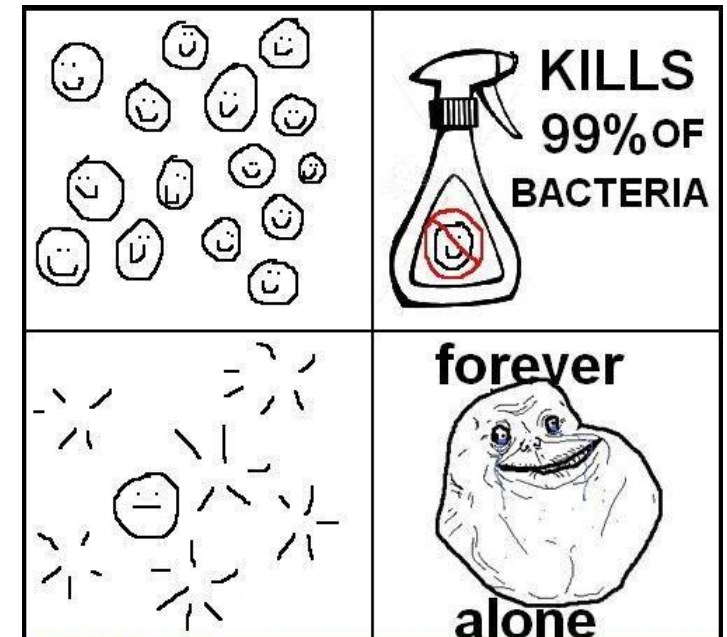
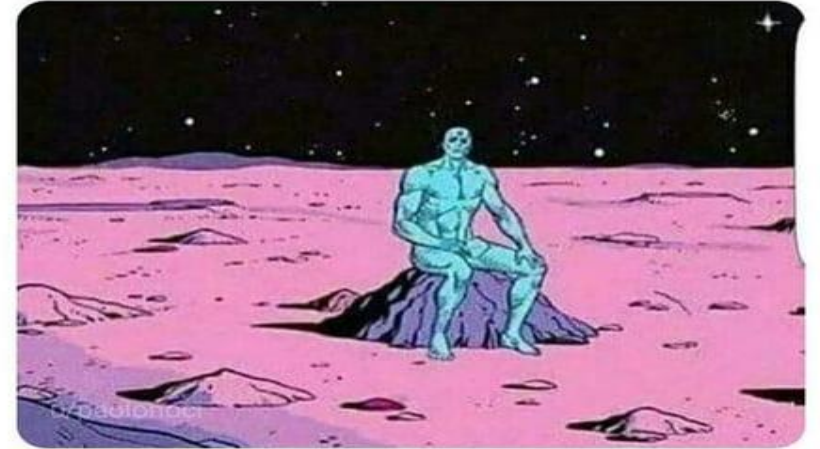
TO TELL ALL THE OTHER GERMS WHAT HAPPENS
WHEN THEY GO ON YOUR HANDS



Example of Variations

- Thus, if there were a population of bacteria living in temperate waters
- If the water temperature were to be increased by global warming, most of these bacteria would die, but the ***few variants resistant to heat*** would survive and grow further.
- *Variation is thus useful for the survival of species over time may not be for an individual.*

When you're a strong bacteria and a detergent kills 99.9% of your friends



What is the importance of DNA copying in reproduction?

Intext Question
Page 128
NCERT

- The DNA is the information source for making proteins and each specific type of protein leads to a specific type of body design. Thus, it is the DNA molecule that determines the body design of an individual.
- It also helps to generate variations during sexual reproduction. This variation leads to evolution.
- Variations increase the survival chances of an organism in the population and continuation of species over time.

Why is variation beneficial to the species but not necessarily for the individual?

- ❑ Variations are beneficial to the species than individual because sometimes for a species, the environmental conditions change so drastically that their survival becomes difficult.
- ❑ For example, if the temperature of water increases suddenly, then most of the bacteria living in that water would die.
- ❑ Only few variants that are resistant to heat would be able to survive.
- ❑ However, if these variants were not there, then the entire species of bacteria would have been destroyed.
- ❑ Thus, these variants help in the survival of the species. However, all variations are not necessarily beneficial for the individual organisms.

Assertion and Reasoning Questions

In each of the following questions, a statement of assertion is given by the corresponding statement of reason. Of the statements, mark the correct answer as

- a) If both Assertion and Reason are true and reason is the correct explanation of Assertion.
- b) If both the Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
- c) If Assertion is true, but Reason is false
- d) If Assertion is false, but Reason is true

Question 1

Assertion- Variations are the sources of evolution

Reason- Errors in DNA copying are called variations

Answer- Option b

Assertion and Reasoning Questions

Question 2

Assertion- DNA acts as information source

Reason- It helps in making different proteins and cellular machinery of the cell

Answer- Option a

Question 3

Assertion- Reproduction is linked to stability of species

Reason- Reproduction is essential for survival of organisms on earth.

Answer- Option c

Let's Recap





[https://suite.smarttech.com/
student/share/12f3879b-
f825-47fe-8618-
9d4af26915dd](https://suite.smarttech.com/student/share/12f3879b-f825-47fe-8618-9d4af26915dd)



CLASS WORK

NOTE THE IMPORTANT
POINTS AND COMPLETE
DIAGRAMS AS DISCUSSED
IN CLASS



HOME WORK

REVISE THE TOPICS AND
LEARN INTEXT QUESTIONS
DISCUSSED TODAY



Pictures and videos taken from Google Youtube.